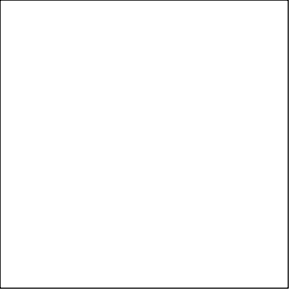
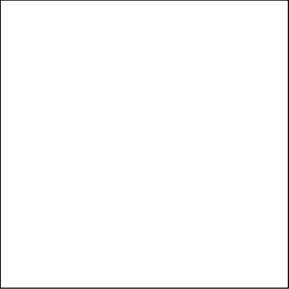


Microcontroller



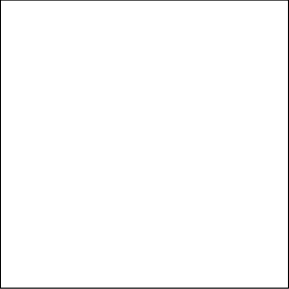
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Sensors



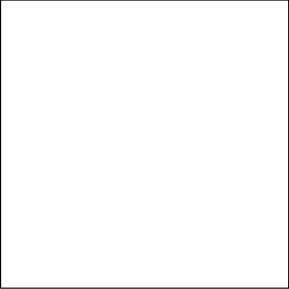
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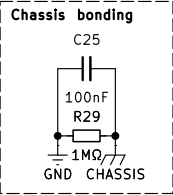


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Power



File: power.kicad_sch

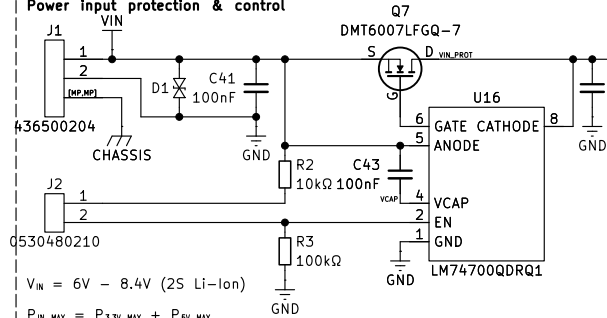


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Power input protection & control

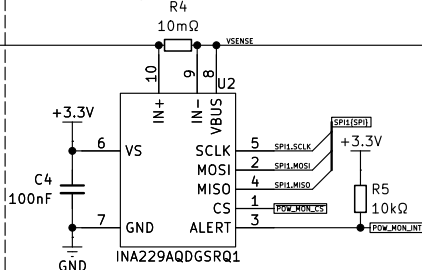


$$V_{IN} = 6V - 8.4V \text{ (2S Li-Ion)}$$

$$P_{IN\ MAX} = P_{3.3V\ MAX} + P_{5V\ MAX} + P_{SENSE\ MAX} + P_{PYRO\ CH\ MAX} = 11.65W + 16.67W + 0.34W + 6.22W = 34.88W$$

$$I_{IN\ MAX} = P_{IN\ MAX} / V_{IN\ MIN} = 34.77W / 6V = 5.81A$$

Power monitoring



$$V_{SENSE\ MAX} = 163.84mV$$

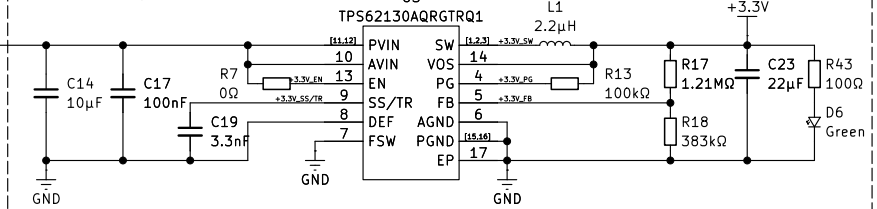
$$R_{SHUNT\ MAX} = V_{SENSE\ MAX} / I_{MAX} = 163.84mV / (34.54W / 6V) = 28.46m\Omega = 10m\Omega$$

$$P_{MAX} = R_{SHUNT} \times I_{MAX}^2 = 10m\Omega \times 5.79A^2 = 0.34W$$

Pyrotechnic channels switch

J3
0530480210

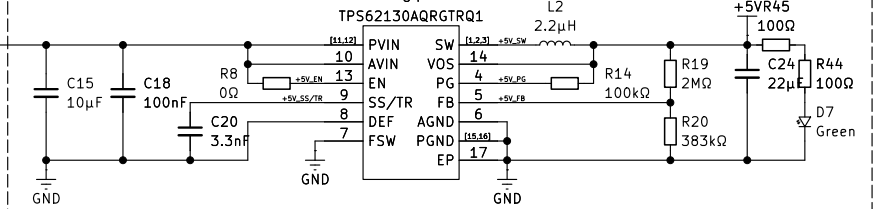
3.3V DC/DC Regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V = (V_{IN} - V_F) / R = ((1.21M\Omega / 383k\Omega) + 1) \times 0.8V = 3.33V$$

$$P_{IN\ MAX} = (P_{3.3V} / \eta_{3.3V} \times 3A) = ((3.3V \times 3A) / 0.85) = 11.65W$$

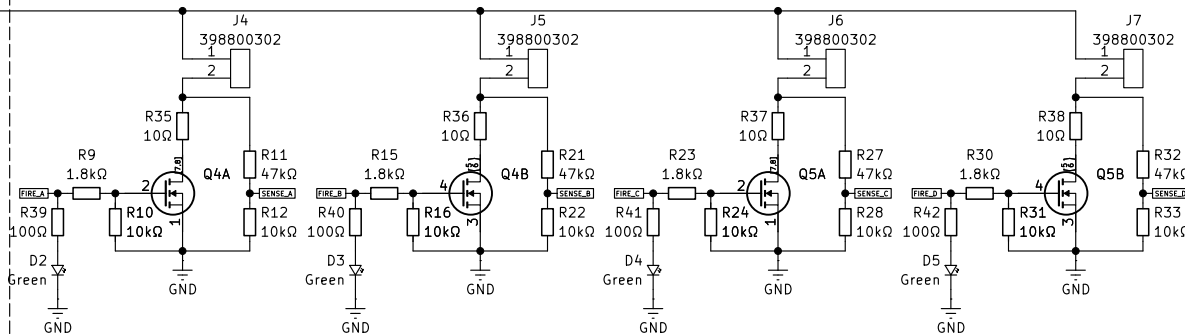
5V DC/DC regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V = (V_{IN} - V_F) / R = ((2M\Omega / 383k\Omega) + 1) \times 0.8V = 5V$$

$$P_{IN\ MAX} = (P_{5V} / \eta_{5V} \times 3A) = ((5V \times 3A) / 0.9) = 16.67W$$

Pyrotechnic channel A,B,C,D



$$R_{LIMIT} = 10\Omega$$

$$t = 100ms$$

$$I_{SENSE\ MAX} = V_{MAX} / R_{TOT} = 8.4V / (47k\Omega + 10k\Omega) = 0.15mA$$

$$I_{LED} = (V_{IN} - V_F) / R = 3.3V - 2.2V / 100\Omega = 11mA$$

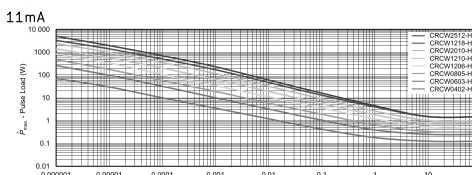
$$I_{MAX\ CH} = V_{MAX} / R_{MIN} + R_{LIMIT} = 8.4V / (1.2\Omega + 10\Omega) = 0.74A$$

$$I_{MIN\ CH} = V_{MIN} / R_{MAX} + R_{LIMIT} = 6V / (47k\Omega + 10k\Omega) = 0.10mA$$

$$P_{LIMIT\ MAX} = R_{LIMIT} \times I_{MAX}^2 = 10\Omega \times 0.74A^2 = 5.48W$$

$$P_{PYRO\ CH\ MAX} = V_{MAX} \times I_{MAX} = 8.4V \times 0.74A = 6.22W$$

Single Pulse



Maximum pulse load, single pulse; applicable if $P \rightarrow 0$ and $n < 1000$ and $Q \leq Q_{MAX}$ for permissible resistance change equivalent to 8000 h operation



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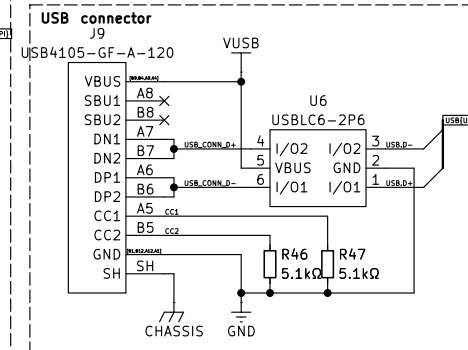
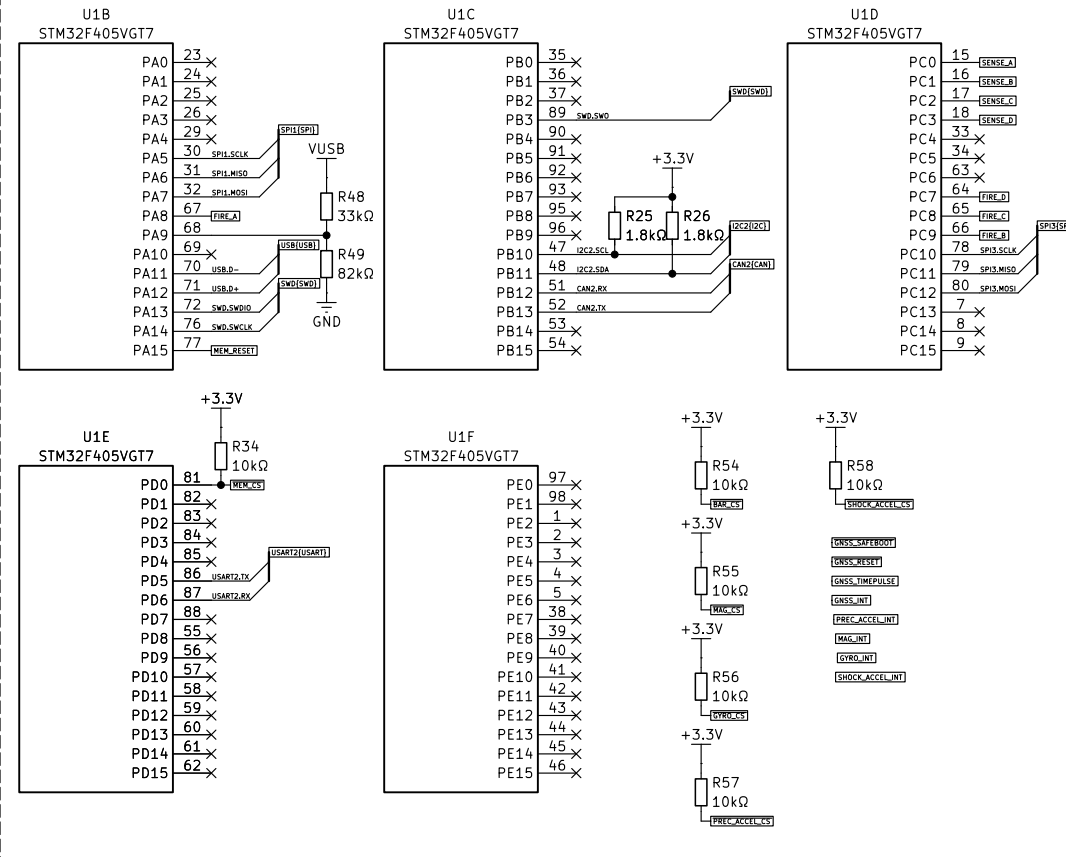
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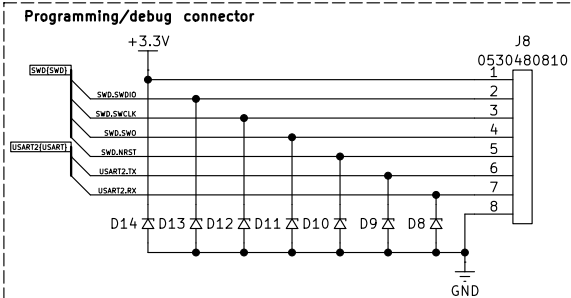
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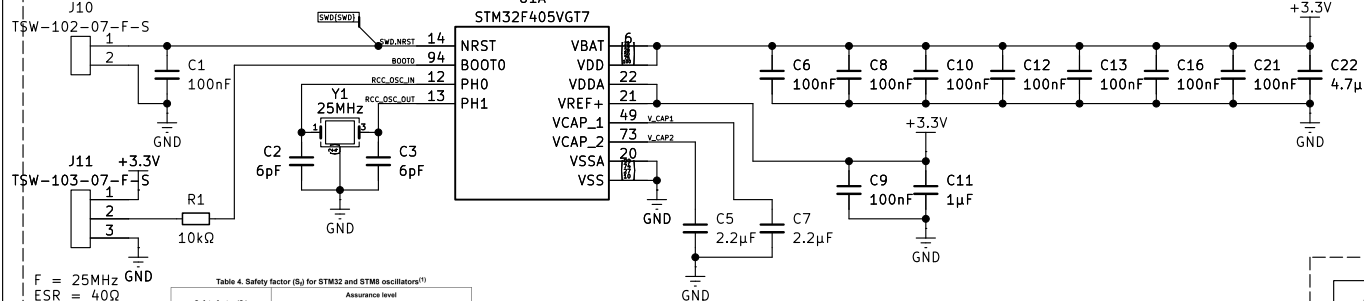
Microcontroller bank A, B, C, D, E



Buzzer
PS124P02BT



Microcontroller core



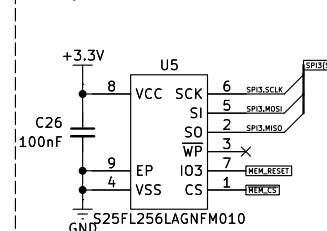
F = 25MHz
ESR = 40Ω
C_L = 8pF
C₀ = 7pF
C_S = 5pF
g_m = 5 mA/V
C_{tot} = C_L + (C_S / 2)

Table 4. Safety factor (S_f) for STM32 and STM8 oscillators⁽¹⁾

Safety factor (S _f)	Assurance level	
	HSE	LSE
S _f ≥ 5	Safe	Very safe
3 ≤ S _f < 5	Not safe	Safe
S _f < 3	Not safe	Not safe

C_{L1/2} = 2 × (C_L - C_S) = 6pF
DL = (ESR × (π × F × C_{tot})² × (V_{DD})²) / 2 = 97.07μW
g_m critical = 4 × (ESR) × (2 × π × F)² × (C₀ + C_L)² = 0.89 mA/V
g_m margin = g_m margin / g_m critical = 5.63

Memory



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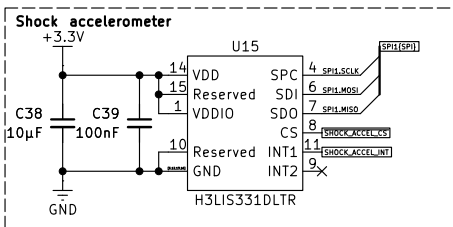
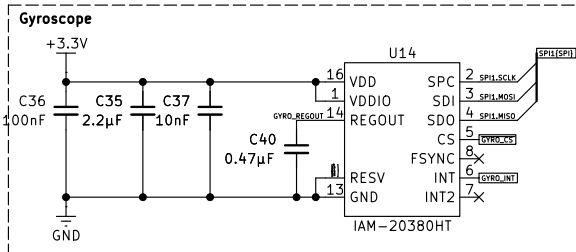
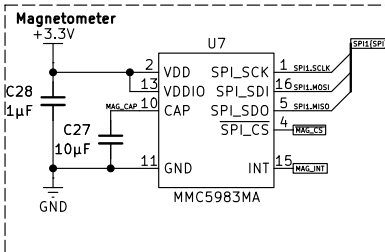
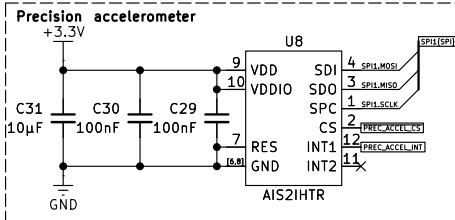
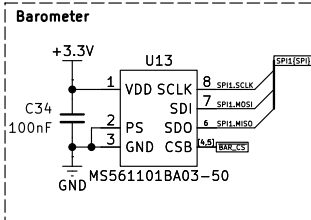
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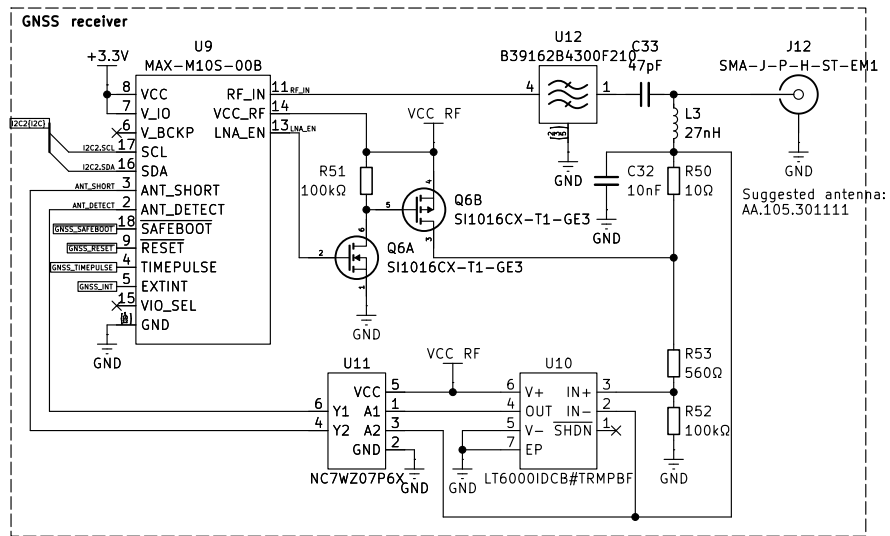


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Radio
CC1200
CC1201



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